NBA Player Statline Forecasting Using Context-Aware Machine Learning

Project Members: Arul Selvakumar

Abstract: Forecasting player performance accurately in professional sports has become essential for both competitive strategy and market-based decision making. In the NBA, predicting player statlines is particularly challenging due to the highly dynamic nature of game environments, including adjustments for player roles, performance trends, and roster availability changes. Conventional approaches that rely on season averages or simple heuristics fail to capture these constantly evolving conditions, leading to inconsistent and unreliable forecasting. This project aims to address this limitation by developing a machine learning framework that produces more accurate and context-aware predictions of player performance.

The project is centered around the construction of a structured ETL pipeline. The first stage involves developing this ETL process using nba_api to gather historical player and team game-to-game statistics. The resulting dataset contains approximately 50,000 player-game observations spanning multiple seasons with over 70 engineered features. These features are constructed to capture key performance drivers, including recent form through rolling averages, player opportunity through minutes-based indicators, and team-level context. By transforming raw game log data into a structured dataset, this stage enables modeling approaches that better reflect real game conditions rather than static historical summaries.

The second stage of the ETL pipeline focuses on model development. Separate supervised learning models are trained for each statistical category—points, rebounds, and assists—using

linear regression, random forest, and gradient boosting techniques. A versioned modeling approach (v0–v2) is implemented to progressively incorporate additional contextual features, allowing for systematic evaluation of feature contributions. Each successive version introduces additional basketball context, enabling direct comparison between baseline models and more expressive approaches. This framework provides insight into how feature engineering influences predictive performance.

The third and final stage focuses on evaluation under realistic forecasting conditions. Models are trained on historical season data and tested on a held-out season (2024–2025) using a time-based split to simulate real-world deployment. Model performance is assessed using MAE and RMSE across all target variables. Results indicate that incorporating contextually engineered features improves predictive performance compared to baseline approaches, and that nonlinear models are better able to capture the complexity of player performance. Context-aware models reduce prediction error by approximately 2-5% compared to baseline approaches, with nonlinear models achieving the most consistent improvements across all target variables.

Practical applicability is demonstrated through a historical replay framework that generates predictions using only information available prior to each game and compares them to actual outcomes. This framework illustrates how the system can function in a real deployment setting where predictions must be made prior to game start.

This project delivers three key outcomes: (1) a reproducible ETL pipeline and structured dataset for NBA player-game analysis, (2) a set of validated machine learning models for context-aware statline forecasting, and (3) a forecasting framework prototype that demonstrates realistic pre-game prediction. These contributions provide a foundation for future extensions, including real-time injury integration and advanced matchup modeling, while already offering value to sports analytics and sports-betting applications seeking more reliable and context-aware performance forecasts.